

Data Set

- Postgres workloads: 217 (20758 samples)
- MySQL workloads: 46 (1314 samples)

Input Features

- **Query Plan Embeddings**
- **Knob Configurations**
- **Workload Features**
- **Hardware Specs:** RAM, CPU cores, and threads
- **DB Type Flag:** 0 (PostgreSQL) or 1 (MySQL)

PostgreSQL Only Results

Algorithm: LightGBM Regressor

Validation: 5-fold GroupKFold Cross-Validation (split by unseen workloads)

Model	CV RMSE	CV MAPE (%)	CV Spearman
LightGBM	0.9710 ± 0.2046	5.54 ± 1.21	0.9417 ± 0.0154
RandomForest	1.0074 ± 0.2921	5.77 ± 1.20	0.9532 ± 0.0171
HistGradientBoosting	1.0429 ± 0.2782	5.85 ± 1.03	0.9244 ± 0.0125

MySQL Adaptation Results (5-fold CV)

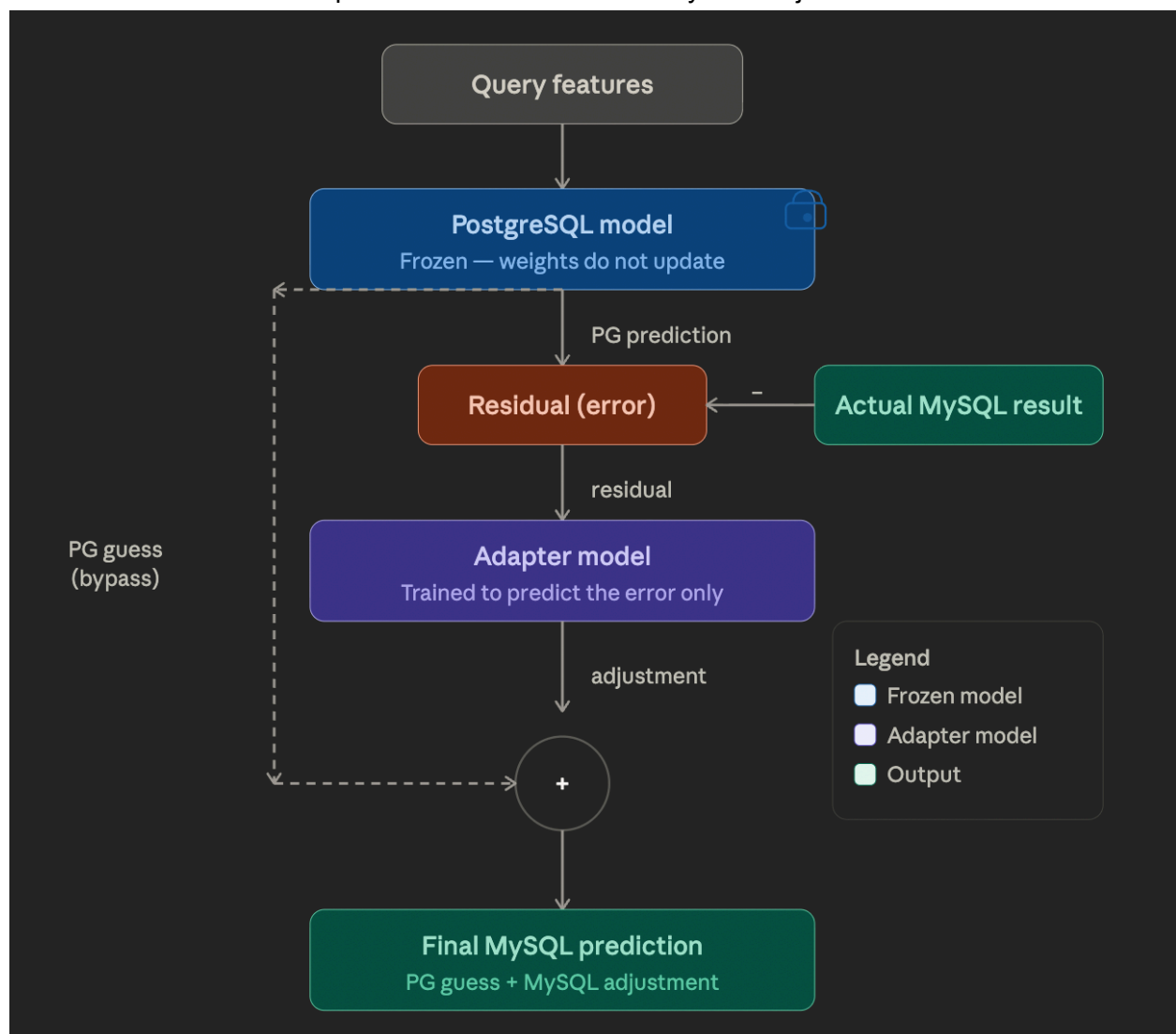
For the adaptation, the best base model (lightgbm) is chosen.

Model Configuration	RMSE	MAPE (%)	Spearman
MySQL-only Regressor	3.3853 ± 0.6655	12.26 ± 5.85	0.4693 ± 0.0437

Model Configuration	RMSE	MAPE (%)	Spearman
Residual Adapter	3.3402 ± 0.6729	5.93 ± 0.73	0.5681 ± 0.0431
Joint weighted training	3.1431 ± 0.5958	4.81 ± 0.23	0.6118 ± 0.0359

Residual Adapter

- Uses a "frozen" PostgreSQL model to predict MySQL performance.
- Identifies the **Residual** (error) between the PG prediction and actual MySQL results.
- Trains a specific adapter model to predict only that error.
- Calculates the final prediction as: PG Guess + MySQL Adjustment.



Joint Weighted Training

- Combines sparse MySQL data (1,314 rows) with the larger PostgreSQL dataset (20,758 rows).
- Assigns higher importance to MySQL rows during training (e.g., a weight of **19.96x**).

Weight = (Number of PG Rows) / (Number of MySQL Rows)

Service / Product	Unit price	Usage	Amount
Primary IPv4	€ 0.0027 per hour	947 Hours	€ 2.56
Server Auction (Pos 2)	€ 0.0529 per hour	440 Hours	€ 23.28
Server Auction (Pos 3)	€ 0.0545 per hour	507 Hours	€ 27.63
	Gross Total (EUR)		€ 53.46
	Tax (0%)		€ 0.00
	Net Total (EUR)		€ 53.46

Service / Product	Unit price	Usage	Amount
Primary IPv4	Rs 1.00 per hour	947 Hours	Rs 954.42
Server Auction (Pos 2)	Rs 19.55 per hour	440 Hours	Rs 8679.25
Server Auction (Pos 3)	Rs 20.15 per hour	507 Hours	Rs 10301.02
	Gross Total (LKR)		Rs 19930.96

Service / Product	Unit price	Usage	Amount
	Tax (0%)		Rs 0.00
	Net Total (LKR)		Rs 19930.96